

Raj Tiwari

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PROFILE SUMMARY

Computer Science undergraduate specializing in **Cyber Security** with strong interest and practical experience in **Generative AI**, **backend engineering**, and **scalable software development**. Experienced in developing **RAG-based systems**, **AI-driven applications**, **REST APIs**, and **cloud-ready deployments** using **Python**, **FastAPI**, **LangChain**, **Hugging Face**, **FAISS**, and **AWS**. Passionate about building intelligent and scalable technology solutions with strong analytical and problem-solving skills.

EXPERIENCE & ACHIEVEMENTS

India Space Lab - Remote

2026

Technical Intern

- Working on technologies related to **Drone Systems**, **CanSat & CubeSat**, **Remote Sensing**, and **GIS** through hands-on technical exposure and guided project activities.
- Gaining practical understanding of **space technologies**, **engineering workflows**, and real-world applications through collaborative learning and industry-oriented tasks.

TECHNICAL PROJECTS

VidSense AI | *Python, FastAPI, LangChain, FAISS, Hugging Face* | [GitHub](#)

2026

- Architected and deployed a production-grade **Retrieval-Augmented Generation (RAG)** system as the primary **AI Developer** for real-time conversational Q&A over YouTube videos, integrating **transcript ingestion**, **semantic chunking**, **dense vector embeddings**, **FAISS similarity search**, and **Hugging Face LLMs**, improving answer grounding and **reducing hallucinations by 40–50%** compared to naive LLM prompting.
- Developed a **memory-aware, multi-turn reasoning pipeline** with **LLM-based query rewriting**, session-level conversational memory, and conditional web search augmentation, enabling robust handling of follow-up questions and incomplete transcripts while maintaining **sub-second retrieval latency** and **top-K context precision >90%** on evaluated queries.
- Implemented **LLMOps** and evaluation workflows using **RAGAS**, tracking context relevance, faithfulness, and answer relevancy; achieved **Context Relevance ≈ 1.0** and **Answer Relevancy $\approx 0.75+$** on benchmark queries, with structured logging, source attribution, and modular model abstraction enabling rapid experimentation across open-source LLMs.

RAGForge | *Automated RAG Optimization & Benchmarking Platform* | [GitHub](#)

2026

- Architected and developed **RAGForge**, an end-to-end RAG optimization and benchmarking platform using **FastAPI**, **PostgreSQL**, **ChromaDB**, **Streamlit**, **Docker**, and **Gemini**, enabling systematic evaluation of retrieval pipelines instead of manual parameter tuning.
- Engineered a configurable benchmarking framework that automatically compares RAG configurations across **embedding models**, **chunk sizes**, **chunk overlaps**, and **top-K retrieval strategies**, generating performance scores, experiment rankings, and leaderboard analytics.
- Implemented a production-style Retrieval-Augmented Generation workflow encompassing **PDF ingestion**, **text extraction**, **semantic chunking**, **vector indexing**, **similarity search**, **LLM-powered answer generation**, and **automated evaluation**.

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript, HTML, CSS

AI & ML: Machine Learning, Deep Learning, NLP, Generative AI, RAG, Prompt Engineering, LLM Evaluation, Model Inference, Vector Databases, Embeddings, Google Gemini

Frameworks & Tools: PyTorch, Scikit-learn, Hugging Face, LangChain, LangGraph, RAGAS, FastAPI, Flask, Docker, MongoDB, FAISS, Chroma, Pandas, NumPy, Streamlit, Git, GitHub, n8n

Cloud & AWS: AWS EC2, S3, VPC, IAM, RDS, Load Balancing, Cloud Architecture, Deployment Automation

Databases: MySQL, MongoDB, Amazon DynamoDB

Core Skills: REST APIs, Backend Development, System Scalability, Data Structures & Algorithms, Problem Solving

EDUCATION

VIT Bhopal University

B.Tech in Computer Science and Engineering (Cyber Security) – CGPA: 8.72

St Mary's Inter College

Higher Secondary: 80.2% Senior Secondary: 95.4%

Aug 2023 – Ongoing

Bhopal, India

2019 – 2022

Etawah, India

CERTIFICATIONS

AWS Certified Solutions Architect – Associate (April 2026) – Validated knowledge of designing secure, scalable, highly available, and cost-optimized AWS architectures using EC2, S3, VPC, IAM, RDS, and Load Balancing.

AWS Certified AI Practitioner (Mar 2026) – AI/ML basics, generative AI, responsible AI, and AWS cloud-based AI services.

Oracle Certified Data Science Professional (Sep 2025) – Cloud infrastructure, data science tools and best practices.